

events backbone (Redis Streams)

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Master technical specification — Volume 3, Control Plane (Go), nano-detail service spec. Deepens the **events backbone** introduced in [02-control-plane §5] (internal/events). This document owns the *byte-level and operational contract* of the event bus: the bus-agnostic Envelope, the five Redis Streams, the full event taxonomy → coordinator reactions, the XReadGroup/XACK/XAUTOCLAIM consumer + dead-letter machinery, at-least-once + idempotent-reaction discipline, the per-event authz rules, the edge cases, the **< 1 s convergence SLO**, and the Phase-2 NATS-JetStream swap behind the same Bus interface. This is a **SPEC** — it describes the implementation; it does not build it. Source evidence cited inline by id: [02_CP] (final/02-control-plane.md), [04_P1] (HelixVPN-Phase1-MVP.md), [04_ARCH §4] (HelixVPN-Architecture-Refined.md §4), [research-go_cp] (cited research, accessed 2026-06-25), [SYNTHESIS §N]. Proto package is unified across the Volume-3 set: **helix.coordinator.v1**.

1. Position, scope, and governing invariants

1.1 What this service owns vs. what it does not

internal/events is the **only path by which a Postgres state mutation reaches the coordinator** [02_CP R3]. It is a thin, durable, bus-agnostic message fabric. It owns:

- the **events.Bus** interface (the swap seam, D3) and its Redis Streams implementation;
- the Envelope wire shape
{id,type,tenant_id,ts,actor,payload,trace_id} [02_CP §5.2];
- the five canonical streams + their consumer-group topology [02_CP §5.1];
- the at-least-once consume loop, XACK discipline, XAUTOCLAIM reclaim, and DLQ routing;
- the idempotency contract every consumer must honour;
- the publish-side transactional-outbox guarantee that couples a DB write to its event.

It does **not** own: the durable tables (those are `internal/store` — the bus carries no truth, only change notifications [02_CP C2]); the topology graph or map-diffing (that is `internal/coordinator` §6, which is *a consumer* of this bus); the agent wire protocol (Coordinator proto §4 — the bus is internal east-west, never exposed to agents).

1.2 Governing invariants this service obeys (from [02_CP §0.1])

#	Invariant	Consequence for the bus
C2	Postgres = truth; Redis is ephemeral . Losing Redis loses no durable state.	A lost event is recoverable by rehydrating the coordinator graph from Postgres (§7.4). The bus is a fast-path, never the system of record.
C3	No-logging by construction.	Event payloads carry identity/topology/policy deltas only — never src/dst IP pairs, ports-in-use, byte counts, flows. A §1.1 payload-lint mirrors the schema-lint [02_CP §2.4] (§9.4).
C4	Default-deny, need-to-know.	Events carry a <code>tenant_id</code> and the coordinator filters fan-out by compiled policy; an event NEVER instructs delivery of a peer a node may not reach.
C5	Push, don't poll; convergence p99 < 1 s .	The bus latency budget (event-publish → consumer-receive) is the largest single term in the SLO; §8 allots it ≤ 250 ms p99.
C7	Package boundaries == service boundaries.	<code>events.Bus</code> is bus-agnostic so the Phase-2 NATS swap is a transport change, not a rewrite (§10).

Decision D3 (restated, [02_CP §1.3, SYNTHESIS §3]). Redis Streams for the MVP, NATS JetStream for Phase-2 scale. Redis is already mandated for presence [05_YBO], so the MVP reuses it with zero new operational dependency; the Bus interface is the seam.

2. The bus-agnostic contract

2.1 `events.Bus` interface (the swap seam)

```
// internal/events/iface.go
//
// Bus is the bus-agnostic seam (D3). The MVP impl is Redis
// Streams (§4); the Phase-2 impl
// is NATS JetStream (§10). NOTHING above this interface knows
// which transport is live.
package events

import (
    "context"
    "time"
```

```

    "github.com/google/uuid"
)

// Stream is a typed stream name. The five canonical values are
// declared in §3.1.
type Stream string

// Bus is the publish/subscribe/ack contract. All methods are
// context-cancellable.
type Bus interface {
    // Publish appends env to stream and returns the transport-
    // assigned message id
    // (Redis stream id "<ms>-<seq>"; NATS sequence as decimal
    // string in P2). At-least-once.
    // MUST be called inside the same logical unit of work as the
    // DB write that caused it
    // (transactional outbox, §5.2) – never best-effort fire-and-
    // forget.
    Publish(ctx context.Context, stream Stream, env Envelope)
        (msgID string, err error)

    // Subscribe joins (group,consumer) to the given streams and
    // returns a delivery channel.
    // The channel yields Delivery values until ctx is cancelled
    // or the bus is closed.
    // Delivery semantics: at-least-once; the caller MUST Ack each
    // Delivery after a
    // successful, idempotent reaction (§6). Redelivery on crash
    // is the contract, not a bug.
    Subscribe(ctx context.Context, group, consumer string,
        streams ...Stream) (<-chan Delivery, error)

    // Ack marks one delivered message as processed for
    // (stream,group). Idempotent: acking an
    // already-acked or unknown id is a no-op, not an error.
    Ack(ctx context.Context, stream Stream, group, msgID string)
        error

    // Nack signals the reaction failed and the message should be
    // retried later. The Redis impl
    // simply does NOT ack (the entry stays in the PEL and is
    // reclaimed by XAUTOCLAIM, §6.4);
    // Nack increments an in-process metric so retries are
    // observable. Provided so callers do
    // not rely on "forget to ack" as the failure signal.
    Nack(ctx context.Context, stream Stream, group, msgID string,
        cause error) error

    // Health returns nil when the transport is reachable (PING
    // ok) AND no consumer group is
    // lagging beyond the alarm threshold (§8.4). Wired into /
    // healthz readiness.
    Health(ctx context.Context) error
}

```

```

    // Close stops all subscriptions and releases the connection
    pool.
    Close() error
}

// Delivery is one received message plus the addressing needed to
    Ack it.
type Delivery struct {
    Stream    Stream
    Group     string
    MsgID     string    // transport message id; the Ack key
    Env       Envelope // decoded payload (§2.2)
    Attempt   int        // 1 on first delivery; >1 means a prior
                        // delivery was not acked (PEL)
    IdleSince time.Time // when this entry was last delivered/
                        // claimed (for XAUTOCLAIM forensics)
}

```

2.2 Envelope — the wire shape (bus-agnostic, [02_CP §5.2])

The envelope is identical across Redis (MVP) and NATS (P2). It is encoded as a **flat Redis hash field-set** on XADD (one field per envelope key; payload is a JSON string) and as a **JSON message body** on NATS. The canonical Go type and its JSON form:

```

// internal/events/envelope.go
package events

import (
    "encoding/json"
    "fmt"
    "time"

    "github.com/google/uuid"
)

type Envelope struct {
    // ID is the transport-assigned message id, filled by the bus
    // on receive (empty on publish).
    // On Redis this is the stream entry id "<ms>--<seq>"; it
    // doubles as the natural dedup key (§6.2).
    ID      string    `json:"id"`
    Type    EventType `json:"type"` // closed vocabulary, §3.3
    TenantID uuid.UUID  `json:"tenant_id"` // RLS/
    // fan-out scope (C4); NEVER empty except SystemTenant
    TS      time.Time `json:"ts"` //
    // RFC3339 UTC, producer clock at publish
    Actor   string    `json:"actor"` //
    // "user:<uuid>" | "device:<uuid>" | "system"
    Payload json.RawMessage `json:"payload"` // type-
    // specific, schema per §3.4; identity/topology/policy ONLY
    // (C3)
}

```

```

    TraceID string          `json:"trace_id"`           // W3C
                    traceparent-derived; correlates DB→bus→stream.Send
    // SchemaV is the payload schema version (default 1). Lets a
    // consumer reject/upgrade an
    // unknown payload shape instead of mis-parsing it (forward-
    // compat, §10.3).
    SchemaV int             `json:"schema_v"`
}

// redisFields renders the envelope as the field map XADD stores.
// payload + ts are stringified.
func (e Envelope) redisFields() map[string]any {
    return map[string]any{
        "type":      string(e.Type),
        "tenant_id": e.TenantID.String(),
        "ts":        e.TS.UTC().Format(time.RFC3339Nano),
        "actor":     e.Actor,
        "payload":   string(e.Payload), // already-compact JSON
        "trace_id":  e.TraceID,
        "schema_v":  e.SchemaV,
    }
}

// New builds an envelope with TS=now, SchemaV=1, and a validated,
// compacted payload.
// It is the ONLY constructor producers use; it enforces the C3
// payload shape via the
// payload lint (§9.4) at build time in tests and at publish time
// as a cheap guard.
func New(t EventType, tenant uuid.UUID, actor string, payload
    any) (Envelope, error) {
    raw, err := json.Marshal(payload)
    if err != nil {
        return Envelope{}, fmt.Errorf("events.New marshal %s:
        %w", t, err)
    }
    if !t.Valid() { // closed-vocabulary guard (§3.3)
        return Envelope{}, fmt.Errorf("events.New: unknown type
        %q", t)
    }
    return Envelope{
        Type: t, TenantID: tenant, TS: time.Now().UTC(),
        Actor: actor, Payload: raw, TraceID:
        traceIDFromContext(), SchemaV: 1,
    }, nil
}

```

Canonical JSON instance (the device.revoked example, [02_CP §5.2]):

```

{
  "id": "1735689600000-0",
  "type": "device.revoked",
  "tenant_id": "8a1f9c2e-0b3d-4e5a-9f6b-1c2d3e4f5a6b",
  "ts": "2026-06-25T12:00:00.000Z",
  "actor": "user:3f2a...",

```

```

"payload": { "device_id": "d41d8cd9-8f00-4b20-
             a8d1-3c2e1f0a9b8c" },
"trace_id":
    "00-4bf92f3577b34da6a3ce929d0e0e4736-00f067aa0ba902b7-01",
"schema_v": 1
}

```

Privacy boundary (C3, anti-bluff). The payload object is structurally constrained to identity/topology/policy keys. There is **no** field anywhere in the taxonomy for a source/destination address pair in use, a port a flow used, a byte count, or a DNS query. advertised_prefixes CIDRs and a device's stable overlay_ip appear (they are topology, not traffic) — exactly the same allowlist the schema-lint enforces [02_CP §2.4]. The §9.4 payload-lint is the runtime signature that proves this is live, not merely promised.

3. Streams, groups, and the event taxonomy

3.1 The five canonical streams + consumer groups [02_CP §5.1]

```

// internal/events/streams.go
const (
    StreamDevices Stream = "events:devices" // identity +
        registry produce
    StreamRoutes Stream =
        "events:routes" // registry produces (connector
        prefixes)
    StreamPolicy Stream = "events:policy" // policy produces
    StreamPresence Stream = "events:presence" // api (heartbeats)
        produces; high-churn, short TTL
    StreamGateway Stream = "events:gateway" // edge health
        probes produce
)

// dlqOf returns the dead-letter stream name for a primary stream
// (§6.5).
func dlqOf(s Stream) Stream { return s + ":dlq" }

```

Stream	Producers	Consumer groups (<group>)	Trim policy (§8.5)
events:devices	identity, registry	coordinator, telemetry, audit	MAXLEN ~ 100000
events:routes	registry	coordinator, telemetry	MAXLEN ~ 50000
events:policy	policy	coordinator, audit	MAXLEN ~ 20000
events:presence	api (heartbeats)	coordinator	MAXLEN ~ 200000 (high churn)
events:gateway	edge health probes	coordinator, telemetry	MAXLEN ~ 50000

Each (stream, group) pair is a durable cursor; multiple helixd replicas in Phase 2 join the **same** group with distinct consumer names so work is partitioned, not duplicated [research-go_cp §5]. In the MVP single-binary deploy there is one consumer per group named by the host id (hostID, the container's stable name).

3.2 Stream/group topology diagram

```

flowchart LR
    subgraph Producers
        ID[identity]
        RG[registry]
        PO[policy]
        AP[api / heartbeats]
        EH[edge health probes]
    end
    ID --> SD[events:devices]
    RG --> SD
    RG --> SR[events:routes]
    PO --> SP[events:policy]
    AP --> SE[events:presence]
    EH --> SG[events:gateway]

    subgraph Groups
        CO[coordinator]
        TE[telemetry]
        AU[audit]
    end
    SD -->|XReadGroup| CO
    SD -->|XReadGroup| TE
    SD -->|XReadGroup| AU
    SR --> CO
    SR --> TE
    SP --> CO
    SP --> AU
    SE --> CO
    SG --> CO
    SG --> TE

    CO -. XAUTOCLAIM reclaim .-> CO
    SD -. poison >5 .-> DLQ[events*:dlq]
    SP -. poison >5 .-> DLQ

```

3.3 EventType — the closed vocabulary

```

// internal/events/types.go
type EventType string

const (
    EvDeviceEnrolled      EventType = "device.enrolled"
    EvDeviceOnline        EventType = "device.online"
    EvDeviceOffline       EventType = "device.offline"
    EvDeviceRevoked       EventType = "device.revoked"

```

```

    EvConnectorAttached      EventType = "connector.attached"
    EvConnectorPrefixesChng  EventType =
        "connector.prefixes.changed"
    EvRouteConflictDetected  EventType = "route.conflict.detected"
    EvPolicyUpdated          EventType = "policy.updated"
    EvPolicyCompiled         EventType = "policy.compiled"
    EvGatewayFailover        EventType = "gateway.failover"
)

var validTypes = map[EventType]Stream{
    EvDeviceEnrolled:      StreamDevices,
    EvDeviceOnline:        StreamPresence, // online/offline are presence, not
        durable identity
    EvDeviceOffline:       StreamPresence,
    EvDeviceRevoked:       StreamDevices,
    EvConnectorAttached:   StreamDevices,
    EvConnectorPrefixesChng: StreamRoutes,
    EvRouteConflictDetected: StreamRoutes,
    EvPolicyUpdated:       StreamPolicy,
    EvPolicyCompiled:      StreamPolicy,
    EvGatewayFailover:     StreamGateway,
}

func (t EventType) Valid() bool { _, ok := validTypes[t]; return
    ok }

// Stream returns the canonical stream a type belongs on.
// Publishing a type to the wrong
// stream is a programmer error caught by a unit test (§9.1) and
// by Publish (§5.2).
func (t EventType) Stream() Stream { return validTypes[t] }

```

3.4 Full taxonomy: payload schema → coordinator reaction → idempotency key

Every payload is {...}-JSON under the envelope's payload. The **Idempotency key** column is the natural key a consumer derives to make the reaction safe under at-least-once redelivery (§6.2). All reactions are recomputed from current graph state, so replay is harmless [02_CP §5.4].

Type	Payload schema (Go struct)	Coordinator reaction	Idempotency key
device.enrolled	{device_id uuid, kind "client" "connector", overlay_ip string}	Insert node into tenant graph; if any active policy grants it, recompute + push upsert-peer delta to grantee maps (C4).	(tenant, device_id)
device.online	{device_id uuid, rtt_ms uint32}	Flip presence→online; mark relay availability; push presence-only delta	(tenant, last-w

Type	Payload schema (Go struct)	Coordinator reaction	Idempotent
device.offline	{device_id uuid}	to peers that already see it. No topology change. Flip presence→offline; peers see relay unavailable; no peer removal (still authorized, just unreachable).	LWW order
device.revoked	{device_id uuid}	Remove node from graph; push remove-peer delta to everyone who could see it ; edge drops the WG peer (< 1 s, §8). Cert marked revoked by pki.	(tenant device, monotonic)
connector.attached	{device_id uuid, site string}	Register connector; allocate site_id for 4via6 (D4); recompute routes for nodes whose policy includes its prefixes.	(tenant device, monotonic)
connector.prefixes.changed	{connector_id uuid, cidrs []string, gen int64}	Recompute routes; push delta to nodes whose compiled policy grants any of these CIDRs (minimal affected set, §6.4). gen is a per-connector monotonic counter — drop a stale gen.	(tenant gen)
route.conflict.detected	{cidr string, connector_ids []uuid}	Flag overlapping-CIDR; surface in Console via events:*→WS (§3.6). Does not block; resolved by 4via6 or operator choice [02_CP §7.3].	(tenant sorted)
policy.updated	{version int64}	(Informational.) Trigger compile if not already dry-run-compiled. Usually a no-op for the coordinator (it waits for policy.compiled).	(tenant version)
policy.compiled	{version int64}	Recompute the whole tenant's visibility from the new CompiledPolicy; diff against last pushed map per node; push minimal deltas. The heaviest reaction.	(tenant version, ignore already applied)

Type	Payload schema (Go struct)	Coordinator reaction	Idempotent
gateway.failover	{from string, to string}	Re-point affected nodes' GatewayInfo.endpoint; push GatewayInfo-only delta.	(tenant LWW)

```
// internal/events/payloads.go – the typed payload structs (one
// per type), all snake_case JSON.
type DeviceEnrolled struct { DeviceID uuid.UUID
    `json:"device_id"`; Kind string `json:"kind"`; OverlayIP
    string `json:"overlay_ip"` }
type DevicePresence struct { DeviceID uuid.UUID
    `json:"device_id"`; RTTMs uint32
    `json:"rtt_ms,omitempty"` }
type DeviceRevoked struct { DeviceID uuid.UUID
    `json:"device_id"` }
type ConnectorAttached struct { DeviceID uuid.UUID
    `json:"device_id"`; Site string `json:"site"` }
type ConnectorPrefixes struct { ConnectorID uuid.UUID
    `json:"connector_id"`; CIDs []string `json:"cids"`; Gen
    int64 `json:"gen"` }
type RouteConflict struct { CIDR string `json:"cidr"`;
    ConnectorIDs []uuid.UUID `json:"connector_ids"` }
type PolicyVersion struct { Version int64
    `json:"version"` }
type GatewayFailover struct { From string `json:"from"`; To
    string `json:"to"` }
```

3.5 Per-event authorization rules (who may emit what)

The bus is internal — agents NEVER publish. Authz is enforced at the *producing module's* public boundary (REST RBAC [02_CP §8.1] / agent mTLS [02_CP §8.2]) **before** events.New. The bus itself enforces two structural rules at Publish:

Event	Permitted producer module	Permitted actor origin	Bus-enforced structural rule
device.enrolled	identity	the enrolled device:<id> via a valid single-use enroll token	tenant_id must match the device's tenant; actor prefix device: or system
device.online/offline	api (heartbeat handler)	device:<id> (mTLS cert)	actor device:<id> must equal payload device_id
device.revoked	identity / api	user:<uuid> with admin role, or system	actor must be user: (admin) or system; never device: (a device)

Event	Permitted producer module	Permitted actor origin	Bus-enforced structural rule
			cannot revoke itself or others)
connector.attached	registry	device:<id> (connector mTLS)	kind=connector asserted from the device row
connector.prefixes.changed	registry	the connector device:<id> or user: admin/operator	prefixes must already be persisted+validated (no unvalidated CIDR on the bus)
route.conflict.detected	registry (compiler dry-run)	system	actor system only
policy.updated/ policy.compiled	policy	user: admin/operator or system	actor must be user:(admin/operator) or system
gateway.failover	(edge health subsystem)	system	actor system only

// internal/events/authz.go – structural guards applied inside the Redis Publish (§5.2).

// These are belt-and-suspenders: the producing module already authorized the action; the bus

// refuses a malformed actor/tenant so a coding bug cannot inject a spoofed-origin event.

```
func guardPublish(stream Stream, env Envelope) error {
    if env.Type.Stream() != stream {
        return fmt.Errorf("events: type %s may not publish to %s", env.Type, stream)
    }
    if env.TenantID == uuid.Nil && env.Actor != "system" {
        return fmt.Errorf("events: non-system event %s requires tenant_id", env.Type)
    }
    switch env.Type {
    case EvDeviceRevoked, EvPolicyUpdated, EvPolicyCompiled:
        if !strings.HasPrefix(env.Actor, "user:") && env.Actor != "system" {
            return fmt.Errorf("events: %s requires user/system actor, got %q", env.Type, env.Actor)
        }
    case EvRouteConflictDetected, EvGatewayFailover:
        if env.Actor != "system" {
            return fmt.Errorf("events: %s requires system actor", env.Type)
        }
    }
}
```

```

    return nil
}

```

3.6 Bridge to the live-UI stream

A subset of events (`device.online`, `route.conflict.detected`, `connector.prefixes.changed`, `gateway.failover`) is re-emitted to the Console WS/SSE endpoint `GET /v1/stream [02_CP §8]`. The `api` module runs a thin `events:* consumer` in group `ws-fanout` that maps an `Envelope` → a UI event JSON and writes it to connected admin sockets. This is read-only projection; it never acks on behalf of coordinator (distinct group), so a slow Console cannot stall map convergence (back-pressure isolation, §6.6).

4. Redis Streams implementation

4.1 Construction & client

Go client: **redis/go-redis/v9** (`XAdd`, `XReadGroup`, `XAck`, `XAutoClaim`, `XPending`, `XTrim` with `MAXLEN`) [research-go_cp §5]. Redis pinned to a known version; if ≥ 8.4 the single-shot `XREADGROUP ... CLAIM` form is available (§6.4 note) — the MVP uses the portable `XAUTOCLAIM`-then-`XREADGROUP` two-step so it does not require 8.4 [research-go_cp §5].

```

// internal/events/redisbus.go
type RedisBus struct {
    rdb      *redis.Client
    hostID string                // stable consumer name
           (container name)
    dlqCap int                  // poison redelivery ceiling
           (default 5, §6.5)
    log      *slog.Logger
    mx      metrics                // §8 counters/histograms
}

func NewRedisBus(rdb *redis.Client, hostID string, log
    *slog.Logger, mx metrics) *RedisBus {
    return &RedisBus{rdb: rdb, hostID: hostID, dlqCap: 5, log:
        log, mx: mx}
}

// EnsureGroups is called once at boot. It creates each
// (stream,group) with MKSTREAM so the
// stream exists even before the first XADD, and starts the cursor
// at "$" (only-new) for a
// FRESH group or "0" to replay backlog – the coordinator uses "0"
// exactly once on cold boot
// AFTER it has hydrated from Postgres, so it never double-applies
// (§7.4).
func (b *RedisBus) EnsureGroups(ctx context.Context, plan
    map[Stream][]string) error {
    for stream, groups := range plan {

```

```

    for _, g := range groups {
        err := b.rdb.XGroupCreateMkStream(ctx,
            string(stream), g, "$").Err()
        if err != nil && !strings.Contains(err.Error(),
            "BUSYGROUP") {
            return fmt.Errorf("xgroupcreate %s/%s: %w",
                stream, g, err)
        }
    }
}
return nil
}

```

4.2 Decode / encode

```

// fromRedis reconstructs an Envelope from an XMessage's Values
// (the redisFields map, §2.2).
func fromRedis(stream Stream, group string, m redis.XMessage,
    attempt int, idle time.Time) (Delivery, error) {
    get := func(k string) string { v, _ := m.Values[k].(string);
        return v }
    tenant, err := uuid.Parse(get("tenant_id"))
    if err != nil && get("tenant_id") != "" { // SystemTenant
        events carry no tenant
        return Delivery{}, fmt.Errorf("bad tenant_id on %s: %w",
            m.ID, err)
    }
    ts, _ := time.Parse(time.RFC3339Nano, get("ts"))
    sv, _ := strconv.Atoi(get("schema_v"))
    env := Envelope{
        ID: m.ID, Type: EventType(get("type")), TenantID: tenant,
        TS: ts,
        Actor: get("actor"), Payload:
            json.RawMessage(get("payload")),
        TraceID: get("trace_id"), SchemaV: sv,
    }
    if !env.Type.Valid() {
        return Delivery{}, fmt.Errorf("unknown event type %q on
            %s", env.Type, m.ID)
    }
    return Delivery{Stream: stream, Group: group, MsgID: m.ID,
        Env: env, Attempt: attempt, IdleSince: idle}, nil
}

```

5. Publish side — transactional outbox guarantee

5.1 The hazard and the rule

[02_CP R3] says "every state mutation emits an event." The hazard: if the DB COMMIT succeeds but the XADD is lost (process crash between the two), the coordinator never learns of the change and a node converges to a stale map — a **silent correctness hole**, worse than a duplicate. Conversely, XADD before COMMIT can publish a change that then rolls back.

Rule P1 — outbox-then-relay (at-least-once, never at-most-once). The producing module writes an outbox row in the **same WithTenant transaction** as the state mutation; a relay loop drains the outbox to Redis and deletes the row only after a successful XADD. A crash after commit re-drains on restart (duplicate XADD possible → harmless, idempotent reactions §6.2). A crash before commit rolls back the outbox row too (no phantom event). This is the canonical transactional-outbox pattern; it is the only way to make "DB write ⇒ event" atomic without distributed transactions [research-go_cp §7 async-path; SYNTHESIS §3].

5.2 Outbox table, relay, and Publish

```
-- migrations/NNNN_outbox.sql (tenant-scoped, RLS like every
    table — [02_CP §2.3])
CREATE TABLE event_outbox (
    id          bigint GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
    tenant_id   uuid NOT NULL REFERENCES tenants(id) ON DELETE
                CASCADE,
    stream      text NOT NULL,                -- "events:policy" etc.
    type        text NOT NULL,                -- closed vocabulary
                (§3.3)
    actor       text NOT NULL,
    payload     jsonb NOT NULL,                -- identity/topology/
                policy only (C3) — payload-lint §9.4
    trace_id    text NOT NULL,
    schema_v    int NOT NULL DEFAULT 1,
    created_at  timestampz NOT NULL DEFAULT now(),
    relayed_at  timestampz                    -- NULL = pending; set
                when XADD succeeds, then row is reaped
);
CREATE INDEX event_outbox_pending ON event_outbox (id) WHERE
    relayed_at IS NULL;
ALTER TABLE event_outbox ENABLE ROW LEVEL SECURITY;
ALTER TABLE event_outbox FORCE ROW LEVEL SECURITY;
CREATE POLICY tenant_isolation ON event_outbox
    USING (tenant_id = current_setting('app.tenant_id')::uuid)
    WITH CHECK (tenant_id = current_setting('app.tenant_id')::uuid);

// internal/events/outbox.go
//
// StageInTx writes the outbox row inside the caller's existing
// tenant transaction. The caller
// (policy.Activate, registry.SetPrefixes, identity.Enroll, ...)
// MUST call this within the same
```

```

// q the state mutation used – that is the atomicity guarantee
// (P1).
func StageInTx(ctx context.Context, q *db.Queries, env Envelope)
    error {
    if err := env.Type.requireStream(env.Type.Stream()); err !=
        nil { // §3.3 guard
        return err
    }
    return q.InsertOutbox(ctx, db.InsertOutboxParams{
        TenantID: env.TenantID, Stream:
            string(env.Type.Stream()), Type: string(env.Type),
        Actor: env.Actor, Payload: env.Payload, TraceID:
            env.TraceID, SchemaV: int32(env.SchemaV),
    })
}

// Relay is the single background loop that moves committed outbox
// rows to Redis. One per
// helixd process; a SKIP-LOCKED claim makes it safe to run on
// every replica in Phase 2.
func (b *RedisBus) Relay(ctx context.Context, store *store.Store)
    error {
    t := time.NewTicker(50 * time.Millisecond) // tight:
        // contributes to the §8 publish budget
    defer t.Stop()
    for {
        select {
        case <-ctx.Done():
            return ctx.Err()
        case <-t.C:
            // FOR UPDATE SKIP LOCKED LIMIT 256: each replica
            // grabs a disjoint batch (no dup relay).
            rows, err := store.ClaimOutboxBatch(ctx, 256)
            if err != nil { b.log.Error("outbox claim", "err",
                err); continue }
            for _, r := range rows {
                env := r.ToEnvelope()
                if err := guardPublish(env.Type.Stream(), env);
                err != nil { // §3.5
                    b.log.Error("outbox guard reject", "id",
                        r.ID, "err", err)
                    _ = store.MarkOutboxRelayed(ctx, r.ID) //
                    // poison-out a malformed row; never block the loop
                    b.mx.outboxRejected.Inc()
                    continue
                }
                id, err := b.xadd(ctx, env) // the
                // actual XADD with MAXLEN trim
                if err != nil { b.mx.publishErrors.Inc();
                break } // leave row pending; retry next tick
                _ = store.MarkOutboxRelayed(ctx, r.ID)

                b.mx.published.WithLabelValues(string(env.Type)).Inc()

                b.mx.outboxLagSeconds.Observe(time.Since(r.CreatedAt).Seconds())
            }
        }
    }
}

```

```

        _ = id
    }
}
}

// xadd is the low-level append with the per-stream trim cap
// ($8.5).
func (b *RedisBus) xadd(ctx context.Context, env Envelope)
(string, error) {
    return b.rdb.XAdd(ctx, &redis.XAddArgs{
        Stream: string(env.Type.Stream()),
        MaxLen: trimCapFor(env.Type.Stream()), // approximate (~)
        trim, 0(1) amortized
        Approx: true,
        Values: env.redisFields(),
    }).Result()
}

// Publish (the Bus interface method) is the NON-transactional
// convenience for the rare
// caller that has no DB write to couple (e.g. gateway.failover
// from the edge-health loop,
// actor=system). It stages to outbox in an autonomous tx then
// lets Relay carry it, so even
// the "no DB mutation" path keeps the same at-least-once +
// ordering guarantees.
func (b *RedisBus) Publish(ctx context.Context, stream
Stream, env Envelope) (string, error) {
    if err := guardPublish(stream, env); err != nil { return "",
err }
    err := b.store.WithSystemOrTenant(ctx, env.TenantID, func(q
*db.Queries) error {
        return StageInTx(ctx, q, env)
    })
    return "", err // real msgID is assigned by Relay's XADD;
    callers needing it read the outbox row
}

```

5.3 Why not publish directly inside the tenant tx?

A direct XADD inside the SQL transaction would couple Redis availability to every DB commit (a Redis blip would roll back legitimate identity/policy writes — violating C2 "losing Redis loses no durable state"). The outbox decouples them: **Postgres commits regardless of Redis health**; the relay catches up when Redis returns. This is the C2-correct ordering.

6. Consume side — at-least-once, idempotent, dead-letter

6.1 The consumer loop (coordinator is the canonical consumer)

```
// internal/events/consumer.go
//
// Run is the durable consume loop for one (group). It is started
// once per group per process.
// Order each tick: (1) reclaim stalled work with XAUTOCLAIM, then
// (2) read new work with
// XREADGROUP > – the production-proven order [research-go_cp §5].
// Every successfully reacted
// message is XACKed; a failed reaction is LEFT in the PEL (no
// ack) so the next XAUTOCLAIM
// retries it; a message past the redelivery ceiling is DLQ'd
// (§6.5).
func (b *RedisBus) Run(ctx context.Context, group string, streams
    []Stream,
    react func(ctx context.Context, d Delivery) error) error {
    for {
        select {
        case <-ctx.Done():
            return ctx.Err()
        default:
        }
        // (1) reclaim: per stream, claim PEL entries idle >=
        // MinIdle (30s) into THIS consumer.
        for _, s := range streams {
            b.reclaimAndHandle(ctx, s, group, react)
        }
        // (2) new work: block up to 2s across all streams for
        // this group.
        args := &redis.XReadGroupArgs{
            Group: group, Consumer: b.hostID, Count: 64, Block: 2
            * time.Second,
            Streams: idsForRead(streams), // [stream1,
            stream2, ..., ">", ">", ...]
        }
        res, err := b.rdb.XReadGroup(ctx, args).Result()
        if err == redis.Nil { continue } // timeout, no new
        // messages – loop
        if err != nil { b.mx.readErrors.Inc(); time.Sleep(200 *
            time.Millisecond); continue }
        for _, st := range res {
            for _, m := range st.Messages {
                d, derr := fromRedis(Stream(st.Stream), group, m,
                    1, time.Now())
                if derr != nil { //
                    // undecodable → straight to DLQ (poison)
                    b.toDLQ(ctx, Stream(st.Stream), group, m,
                        derr, 1); continue
                }
            }
        }
    }
}
```

```

    }
    b.handleOne(ctx, group, react, d)
  }
}
}

// handleOne runs the reaction under a per-message timeout, then
// acks on success.
func (b *RedisBus) handleOne(ctx context.Context, group string,
    react func(context.Context, Delivery) error, d Delivery) {

    start := time.Now()
    hctx, cancel := context.WithTimeout(ctx, 3*time.Second)
    defer cancel()
    err := react(hctx, d)
    b.mx.reactSeconds.WithLabelValues(string(d.Env.Type)).Observe(time.Since(start))
    if err == nil {
        if aerr := b.rdb.XAck(ctx, string(d.Stream), group,
            d.MsgID).Err(); aerr != nil {
            b.mx.ackErrors.Inc() // ack failed → entry stays in
            // PEL → reclaimed later (still safe)
        }
        return
    }
    // reaction failed → DO NOT ack. Leave in PEL; XAUTOCLAIM
    // retries; ceiling routes to DLQ.
    b.mx.reactErrors.WithLabelValues(string(d.Env.Type)).Inc()
    b.log.Warn("reaction failed", "type", d.Env.Type, "id",
        d.MsgID, "attempt", d.Attempt, "err", err)
}

```

6.2 Idempotency contract (the heart of at-least-once safety)

At-least-once means **every consumer MUST be safe under redelivery** [02_CP §5.4, research-go_cp §5]. Two cooperating mechanisms:

1. **Recompute-from-state reactions (primary).** The coordinator never applies a *delta of a delta*; it applies the event to its in-memory graph and **recomputes** the affected nodes' maps from current state. Replaying `policy.compiled v=7` twice yields the identical map the second time — a no-op diff, nothing pushed. This makes most reactions naturally idempotent with no dedup store [02_CP §5.4].
2. **Monotonic-version / seen-set guard (for non-idempotent side effects).** Where a reaction has an external side effect that is NOT naturally idempotent (e.g. the audit group writes an `audit_events` row; the `ws-fanout` group emits a UI toast), the consumer derives the §3.4 **Idempotency key** and guards on it:

```

// internal/events/idem.go
//
// The coordinator tracks the last-applied monotonic marker per
// (tenant,kind) in-memory; an

```

```

// event whose version/gen is <= the applied marker is dropped.
// Durable consumers (audit) use a
// small Redis SET with TTL as the seen-set so a duplicate XADD
// does not write two audit rows.
type applied struct {
    policyVersion map[uuid.UUID]int64           // tenant -> last
        applied policy.compiled version
    connectorGen  map[uuid.UUID]int64           // connector ->
        last applied prefixes gen
    revoked       map[uuid.UUID]struct{}        // device ids
        already removed (terminal)
}

// shouldApply returns false (skip) when the event is stale/
// duplicate by its monotonic key.
func (a *applied) shouldApply(d Delivery) bool {
    switch d.Env.Type {
    case EvPolicyCompiled:
        var p PolicyVersion; _ = json.Unmarshal(d.Env.Payload, &p)
        if p.Version <= a.policyVersion[d.Env.TenantID] { return
            false } // already applied or older
        a.policyVersion[d.Env.TenantID] = p.Version
        return true
    case EvConnectorPrefixesChng:
        var c ConnectorPrefixes; _ =
            json.Unmarshal(d.Env.Payload, &c)
        if c.Gen <= a.connectorGen[c.ConnectorID] { return false }
        a.connectorGen[c.ConnectorID] = c.Gen
        return true
    case EvDeviceRevoked:
        var r DeviceRevoked; _ = json.Unmarshal(d.Env.Payload, &r)
        if _, done := a.revoked[r.DeviceID]; done { return
            false } // revoke is terminal+idempotent
        a.revoked[r.DeviceID] = struct{}{}
        return true
    default:
        return true // online/offline/enrolled/etc. are recompute-
            from-state idempotent (mechanism 1)
    }
}

```

Ordering note. Redis Streams preserve per-stream order; the monotonic guard means an out-of-order *retry* of an older version (delivered late by XAUTOCLAIM after a newer one already applied) is correctly dropped, not regressed. This is why the taxonomy keys are monotonic (version, gen) rather than wall-clock for the topology-changing events.

6.3 Consumer state machine (per message)

stateDiagram-v2

```

[*] --> Delivered: XREADGROUP > (or XAUTOCLAIM reclaim)
Delivered --> Decoding
Decoding --> Poisoned: decode error
Decoding --> Reacting: decoded ok

```

```

Reacting --> Skipped: shouldApply()==false (stale/dup)
Reacting --> Reacted: react() ok
Reacting --> Failed: react() error
Skipped --> Acked: XACK
Reacted --> Acked: XACK
Failed --> Pending: NO ack (stays in PEL)
Pending --> Reacting: XAUTOCLAIM (idle>=30s), attempt++
Pending --> Poisoned: attempt > ceiling(5)
Poisoned --> DLQ: XADD events*:dlq + XACK off main
Acked --> [*]
DLQ --> [*]

```

6.4 XAUTOCLAIM reclaim — the crash-recovery path

```

// reclaimAndHandle transfers stalled PEL entries (idle >= 30s) to
// this consumer and re-reacts.
// This is the dead-letter recovery path: a crashed coordinator's
// un-acked entries are NOT lost
// – composes constitution §11.4.147 (no-work-loss) [02_CP §5.4].
func (b *RedisBus) reclaimAndHandle(ctx context.Context, s
    Stream, group string,
    react func(context.Context, Delivery) error) {

    cursor := "0-0"
    for {
        msgs, next, err := b.rdb.XAutoClaim(ctx,
            &redis.XAutoClaimArgs{
                Stream: string(s), Group: group, Consumer: b.hostID,
                MinIdle: 30 * time.Second, Start: cursor, Count: 64,
            }).Result()
        if err != nil { b.mx.claimErrors.Inc(); return }
        for _, m := range msgs {
            // delivery count from XPENDING tells us the attempt;
            // ceiling check first.
            attempt := b.deliveryCount(ctx, s, group, m.ID)
            if attempt > b.dlqCap {
                b.toDLQ(ctx, s, group, m, errPoison, attempt)
                continue
            }
            d, derr := fromRedis(s, group, m, attempt, time.Now())
            if derr != nil { b.toDLQ(ctx, s, group, m, derr,
                attempt); continue }
            b.handleOne(ctx, group, react, d)
        }
        if next == "0-0" || len(msgs) == 0 { return } // cursor
        // wrapped – done this pass
        cursor = next
    }
}

```

Redis 8.4 shortcut (deferred). Redis ≥ 8.4 folds reclaim into the read via XREADGROUP ... CLAIM <min-idle> [research-go_cp §5]. The MVP keeps the portable two-step so it does not pin 8.4; §10.4 lists the one-line swap if the deployment can guarantee ≥ 8.4 .

6.5 Dead-letter routing (poison messages)

```
// toDLQ moves a message past the redelivery ceiling (or an
// undecodable one) to the per-stream
// DLQ and ACKs it off the main PEL so it stops being redelivered.
// The DLQ entry carries the
// delivery count + first-seen + the failure cause for forensics
// [research-go_cp §5].
func (b *RedisBus) toDLQ(ctx context.Context, s Stream, group
    string, m redis.XMessage, cause error, attempt int) {
    dlq := dlqOf(s)
    fields := map[string]any{}
    for k, v := range m.Values { fields[k] = v }
    fields["dlq_reason"] = cause.Error()
    fields["dlq_attempts"] = attempt
    fields["dlq_group"] = group
    fields["dlq_orig_id"] = m.ID
    fields["dlq_at"] = time.Now().UTC().Format(time.RFC3339Nano)
    // DLQ retention OUTLIVES the main stream (audit), so a larger
    // MAXLEN.
    _ = b.rdb.XAdd(ctx, &redis.XAddArgs{Stream: string(dlq),
        MaxLen: 1_000_000, Approx: true, Values: fields}).Err()
    _ = b.rdb.XAck(ctx, string(s), group, m.ID).Err() // remove
    // from main PEL
    b.mx.dlqTotal.WithLabelValues(string(s),
        string(group)).Inc() // helix_events_dlq_total [02_CP
    // §5.4]
    b.log.Error("event dead-lettered", "stream", s, "group",
        group, "id", m.ID,
        "attempts", attempt, "reason", cause)
}
```

A DLQ entry is **never silently dropped**; `helix_events_dlq_total` increments fire an alert (§8.4). The DLQ is inspectable (`XRANGE events:policy:dlq - +`) and replayable by an operator tool (`helixvpctl events replay-dlq <stream>`), which re-XADDS the original fields onto the primary stream after the operator fixes the cause — closing the §11.4.147 no-work-loss loop.

6.6 Back-pressure & slow-consumer isolation

Each consumer group has an independent cursor; a slow audit/ws-fanout/telemetry group **cannot** stall the coordinator group (distinct PELs). Within the coordinator, the **per-stream send-queue to each agent is bounded** [02_CP §6.4]; a slow agent that fills its queue is dropped and forced to reconnect-with-snapshot — so a slow *downstream* never converts into unbounded coordinator memory or a stuck event cursor (the §8 24 h-soak SLO).

7. End-to-end reconciliation through the bus (the < 1 s promise)

7.1 The policy-activate path (event publish → delta on wire)

sequenceDiagram

```
autonumber
participant API as api (Gin)
participant POL as policy
participant PG as Postgres (RLS)
participant OB as event_outbox
participant RL as Relay loop
participant RS as Redis Stream events:policy
participant CO as coordinator (group)
participant ED as Rust edge / agents
API->>POL: Activate(v)
POL->>PG: deactivate all + activate v (one tx)
POL->>OB: StageInTx(policy.compiled, v)  %% same tx - atomic
```

(P1)

```
Note over PG,OB: COMMIT - DB truth + outbox row land together
RL->>OB: ClaimOutboxBatch (FOR UPDATE SKIP LOCKED)
RL->>RS: XADD events:policy {policy.compiled, v} +
```

MarkOutboxRelayed

```
RS-->>CO: XReadGroup(coordinator) delivers
CO->>CO: shouldApply(v) → recompute tenant visibility (minimal
set, §6.4)
```

```
loop each affected open WatchNetworkMap stream
```

```
CO->>ED: stream.Send(MapDelta)
```

```
end
```

```
CO->>RS: XACK
```

Note over API,ED: target p99 < 1s event→enforced, ZERO restarts

7.2 Latency budget decomposition (how < 1 s is met)

Segment	Budget (p99)	Mechanism
COMMIT → outbox row visible	≤ 20 ms	same-tx insert; index event_outbox_pending
Relay tick latency (claim→XADD)	≤ 70 ms	50 ms ticker + SKIP LOCKED batch
Redis XADD→consumer XReadGroup	≤ 60 ms	BLOCK 2s returns immediately on append
coordinator recompute + diff (tenant ≤ 1k devices)	≤ 200 ms	in-mem graph; minimal affected set §6.4; [02_CP §10.2 compile <200ms]
stream.Send enqueue → edge reconcile	≤ 150 ms	bounded per-agent queue; HTTP/2 server-stream
Total event→enforced		

Segment	Budget (p99)	Mechanism
	≤ 500 ms typical, < 1 s p99	measured by <code>helix_reconcile_seconds</code> [02_CP §10.2]

The bus's own contribution (outbox+relay+XADD+deliver) is budgeted ≤ 250 ms p99 — comfortably inside the 1 s SLO, leaving headroom for the coordinator's recompute and the edge apply.

7.3 Revoke path (the < 1 s security-critical case)

`device.revoked` is the latency-critical security event. It travels the same outbox→relay→events:devices→coordinator path; the coordinator removes the node, pushes a remove-peer delta to every map that contained it, and the edge removes the kernel WG peer — all within the same < 1 s budget [02_CP §9.3]. Because revoke is **terminal + idempotent** (§6.2), a redelivered `device.revoked` is a no-op (the node is already gone), so an XAUTOCLAIM retry after a coordinator crash re-enforces rather than errors.

7.4 Cold-boot rehydration (Redis-loss / coordinator-restart correctness, C2)

flowchart TD

```

    A[helixd boot] --> B[hydrate in-mem graph from Postgres<br/>
    >devices, connectors, prefixes, groups, active policy]
    B --> C{Redis stream backlog?}
    C -->|fresh group| D[XGROUPCREATE at '$'<br/>only-new – no
    replay needed,<br/>graph already current from PG]
    C -->|existing group, lag| E[XAUTOCLAIM own PEL +<br/>
    >XREADGROUP '>' to drain backlog<br/>monotonic guard drops stale
    §6.2]
    D --> F[serve WatchNetworkMap streams]
    E --> F
  
```

Because Postgres is truth (C2), the coordinator **always** rebuilds its graph from Postgres on boot, then attaches to the bus for the *delta* stream. A totally lost Redis (flushed, restarted empty) costs nothing durable: the next state mutation re-publishes via the outbox, and any in-flight change that was only on the bus is reconstructable because its DB write already committed (the outbox row either relayed or will re-relay). This is the C2 guarantee made concrete: **the bus is a fast-path accelerator over Postgres truth, never a second source of truth.**

8. SLO, metrics, and operational thresholds

8.1 The convergence SLO (anti-bluff, measured)

Metric	Target	How measured [02_CP §10.2]
event-publish → consumer-receive	$p99 \leq 250$ ms	helix_event_bus_seconds (outbox.created_at → consumer fromRedis)
event → delta-on-wire (whole chain)	p99 < 1 s	helix_reconcile_seconds (event-receive → stream.Send)
device revoke → edge enforced	< 1 s	revoke-to-WG-peer-removed timer on the edge
outbox relay lag	$p99 \leq 100$ ms	helix_event_outbox_lag_seconds (created_at → relayed)
DLQ rate	0 sustained	helix_events_dlq_total (any sustained increase alerts)
coordinator memory @ 10k streams	bounded, no leak / 24 h	process_resident_memory_bytes slope ≈ 0

8.2 Prometheus metric definitions

```
// internal/events/metrics.go
type metrics struct {
    published      *prometheus.CounterVec //
        helix_events_published_total{type}
    publishErrors  prometheus.Counter      //
        helix_events_publish_errors_total
    outboxRejected prometheus.Counter      //
        helix_events_outbox_rejected_total (guard reject)
    outboxLagSeconds prometheus.Histogram    //
        helix_event_outbox_lag_seconds
    busSeconds     *prometheus.HistogramVec //
        helix_event_bus_seconds{type} (publish→receive)
    reactSeconds   *prometheus.HistogramVec //
        helix_event_react_seconds{type}
    reactErrors    *prometheus.CounterVec   //
        helix_event_react_errors_total{type}
    readErrors     prometheus.Counter       //
        helix_events_read_errors_total
    claimErrors    prometheus.Counter       //
        helix_events_claim_errors_total
    ackErrors      prometheus.Counter       //
        helix_events_ack_errors_total
    dlqTotal       *prometheus.CounterVec   //
        helix_events_dlq_total{stream,group}
    pelDepth       *prometheus.GaugeVec     //
        helix_events_pel_depth{stream,group} (lag gauge)
}
```


8.3 Consumer-group lag gauge

A background sampler runs `XPENDING <stream> <group>` every 5 s and sets `helix_events_pel_depth{stream,group}` to the pending-entries count. A growing PEL depth is the leading indicator of a stuck consumer **before** the DLQ rate moves — it feeds the Health readiness check.

8.4 Alert thresholds (operational)

Condition	Threshold	Action
<code>helix_events_dlq_total</code> rate	> 0 over 1 min	page; inspect DLQ, fix cause, <code>replay-dlq</code>
<code>helix_events_pel_depth</code> (coordinator)	> 1000 sustained 30 s	warn; consumer likely stalled/slow
<code>helix_event_bus_seconds</code> p99	> 250 ms 5 min	warn; Redis or relay degraded
<code>helix_event_outbox_lag_seconds</code> p99	> 500 ms 5 min	warn; relay loop starved
<code>Bus.Health</code> PING fail	any	readiness NOT READY (fail-static: existing tunnels keep forwarding, C1)

8.5 Stream trimming (bounded memory)

Each XADD carries the per-stream `MAXLEN ~ N` from the §3.1 table so a stream never grows unbounded [research-go_cp §5]. Presence (`events:presence`) gets the largest cap (high churn); policy the smallest (rare, high-value). DLQ streams use `MAXLEN ~ 1_000_000` and outlive the primary for audit. A nightly `XTRIM MINID` reaps anything older than 24 h on the non-DLQ streams as a backstop.

9. Test points (constitution §11.4.169 — comprehensive test-type coverage)

§11.4.169 mandates comprehensive test-type coverage; every claim below is backed by a concrete, anti-bluff test with captured evidence per §11.4.5/§11.4.69/§11.4.107. Mocks/stubs are permitted **only** in unit tests (§11.4.27); all other types exercise a real Redis + Postgres booted on-demand via the `vasic-digital/containers` submodule (§11.4.76) — never ad-hoc `docker run`.

9.1 Unit tests (mocks allowed)

- **Envelope round-trip:** `New` → `redisFields` → `fromRedis` is byte-stable for every type; payload JSON is canonical/compact; `schema_v` defaults to 1.

- **Type/stream mapping:** `EventType.Stream()` returns the §3.3 table for all 10 types; `guardPublish` rejects a type on the wrong stream and a non-user/system actor on `device.revoked/policy.*`.
- **Idempotency guard (`shouldApply`):** replaying `policy.compiled v=7` twice → second call returns `false`; an out-of-order `v=6` after `v=7` → `false`; `connector.prefixes.changed` stale gen → `false`; duplicate `device.revoked` → `false` after first.

9.2 Integration tests (real Redis + Postgres, §11.4.76)

- **Transactional outbox atomicity (P1):** kill the relay between COMMIT and XADD; restart; assert the event still reaches the stream exactly once acked (at-least-once, no loss, no phantom on a rolled-back tx). Captured evidence: stream X RANGE + outbox row state.
- **At-least-once + idempotent reaction:** force a redelivery (don't ack, wait `MinIdle`, `XAUTOCLAIM`); assert the coordinator graph + pushed deltas are identical to the single-delivery run (recompute-from-state idempotence, §6.2).
- **DLQ routing:** inject a poison `policy.compiled` whose reaction always errors; assert it is redelivered exactly `ceiling=5` times then lands in `events:policy:dlq` with `dlq_attempts=6`
 - `dlq_reason`, and `helix_events_dlq_total` increments; assert it is gone from the main PEL.
- **Cold-boot rehydration (C2):** flush Redis; restart `helixd`; assert the graph is rebuilt from Postgres and `WatchNetworkMap` serves a correct snapshot with zero durable loss.

9.3 End-to-end / full-automation tests

- Drive `enroll` → `connector.attached` → `connector.prefixes.changed` → `policy.compiled` through the real bus; assert the `WatchNetworkMap` delta stream content **and** the event → delta `helix_reconcile_seconds p99 < 1 s` with captured evidence (§11.4.69/§11.4.107) — the same SLO the parent doc asserts [02_CP §10.3].
- Revoke E2E: `device.revoked` → edge WG-peer-removed timer < 1 s.

9.4 Payload no-logging lint (C3) + paired §1.1 mutation

```
// tools/payloadlint/main.go – parses internal/events/payloads.go
// + every events.New call site
// and FAILS the build if any payload struct/field matches the
// forbidden traffic-log shape
// (mirrors the schema-lint [02_CP §2.4]). Allowlist: device_id,
// overlay_ip, cidr(s),
// connector_id, version, gen, site, kind, rtt_ms, from, to –
// identity/topology/policy only.
var forbiddenPayloadFields = regexp.MustCompile(
    `(?!i)\b(src_ip|dst_ip|dest_ip|src_port|dst_port|bytes_(in|
    out)|packet_count|payload_bytes|sni_host|flow_id|
    dns_query)\b`)
```

Paired §1.1 mutation: a test that adds a `BytesOut int json:"bytes_out"` field to a payload struct MUST make `payloadlint` FAIL; removing it MUST make it pass. This is the runtime signature (§11.4.108) proving the bus carries no traffic data — C3 is live, not promised.

9.5 Stress + chaos (§11.4.85)

- **Stress:** 50k events across the five streams at 5k/s; assert no PEL growth beyond the §8.4 threshold, `helix_event_bus_seconds p99` $\leq$ 250 ms, zero DLQ.
- **Chaos — consumer kill:** SIGKILL the coordinator mid-batch; assert XAUTOCLAIM reclaims the un-acked PEL on restart and no event is lost (§11.4.147 no-work-loss).
- **Chaos — Redis partition:** drop the Redis connection for 30 s; assert Postgres commits keep succeeding (C2), the outbox accumulates, and the relay drains the backlog on reconnect with no duplicate-induced incorrect map (idempotent reactions absorb the duplicates).
- **Chaos — clock skew:** publish events with a producer clock 5 s ahead; assert the monotonic version/gen guards (not wall-clock) keep ordering correct.

10. Phase-2 NATS-JetStream swap (the bus-agnostic seam, D3)

10.1 Why the swap is a transport change, not a rewrite

The `events.Bus` interface (§2.1) and the `Envelope` (§2.2) are the entire surface above the transport. NATS JetStream provides the same primitives the Redis impl uses [SYNTHESIS §3, 04_P1 §12]:

Redis Streams (MVP)	NATS JetStream (P2)	Maps to
stream <code>events:policy</code>	JetStream stream EVENTS_POLICY (subject <code>events.policy.*</code>)	Stream name
consumer group coordinator	durable pull consumer coordinator	<code>Subscribe(group,...)</code>
XReadGroup >	<code>fetch()</code> on the pull consumer	delivery channel
XACK	<code>msg.Ack()</code>	<code>Bus.Ack</code>
XAUTOCLAIM (MinIdle reclaim)	JetStream <code>AckWait</code> + auto- redelivery	reclaim (built-in)
<code>:dlq stream + ceiling</code>	<code>MaxDeliver</code> + advisory → DLQ subject	toDLQ
MAXLEN ~ N	stream <code>MaxMsgs/MaxAge</code> retention	trim
XADD outbox relay	JetStream <code>publish</code> (same outbox relay)	Relay/xadd

10.2 What changes / what does not

Changes (transport only): a new `internal/events/natsbus.go` implementing Bus; the deployment swaps Redis-for-bus → NATS (Redis stays for presence in some topologies, or presence also moves). **Unchanged:** the outbox table + relay (relay's `xadd` becomes a NATS publish), the Envelope, the taxonomy, every consumer's reaction + idempotency guard, the authz guards, the metrics names, and every test in §9 (they target the Bus interface, so they re-run unchanged against the NATS impl — the §11.4.81 cross-platform-parity discipline applied to the transport axis).

10.3 Forward-compat already wired

`Envelope.SchemaV` lets a Phase-2 consumer reject/upgrade an unknown payload shape rather than mis-parse it; the closed `EventType` vocabulary + `validTypes` map mean a new event type is an additive map entry, not a reshaping. NATS multi-region fan-out (the actual reason to swap) needs no envelope change — `tenant_id` already scopes routing.

10.4 The one-line MVP optimisation (Redis ≥ 8.4)

If the MVP deployment can guarantee Redis ≥ 8.4, the §6.4 two-step reclaim collapses to the single-shot `XREADGROUP ... CLAIM <min-idle>` [research-go_cp §5] — a localized change inside Run only; the Bus interface and all callers are untouched.

11. Phase → task → subtask plan (events backbone slice)

Derived from [02_CP §13] CP-T5, expanded to nano-detail. Each leaf is a workable item (§11.4.93). Risk-descending (§11.4.132): the outbox atomicity + payload-lint first (they are the correctness/privacy floor).

- **EV-T1 — bus-agnostic contract.**
 - EV-T1.1 `events.Bus` interface + Envelope + `EventType` closed vocabulary + `validTypes`.
 - EV-T1.2 `events.New` constructor + `guardPublish` structural authz (§3.5).
- **EV-T2 — transactional outbox (the atomicity floor).**
 - EV-T2.1 `event_outbox` table + RLS + `event_outbox_pending` index (§5.2).
 - EV-T2.2 `StageInTx` + `ClaimOutboxBatch` (FOR UPDATE SKIP LOCKED) + `MarkOutboxRelayed`.
 - EV-T2.3 Relay loop (50 ms tick) + `xadd` with `MAXLEN ~ trim`; P1 atomicity integration test (§9.2).
- **EV-T3 — Redis Streams impl.**
 - EV-T3.1 `RedisBus` + `EnsureGroups` (MKSTREAM) + `fromRedis/redisFields` codec.
 - EV-T3.2 Run consume loop: `XAUTOCLAIM`-then-`XReadGroup` + `handleOne` + `XACK` (§6.1).

- EV-T3.3 `reclaimAndHandle` (MinIdle 30 s) + delivery-count ceiling + toDLQ + replay tool.
- **EV-T4 — idempotency + reactions.**
 - EV-T4.1 `applied/shouldApply` monotonic guards (policy version, connector gen, revoke).
 - EV-T4.2 coordinator reaction adapter (recompute-from-state, §6.2 mechanism 1).
- **EV-T5 — observability + SLO.**
 - EV-T5.1 the §8.2 metric set + `helix_event_bus_seconds/helix_reconcile_seconds` wiring.
 - EV-T5.2 PEL-depth sampler (XPENDING) + `Bus.Health` + §8.4 alert rules.
- **EV-T6 — privacy + anti-bluff gates.**
 - EV-T6.1 `tools/payloadlint` (§9.4) + paired §1.1 mutation (add `bytes_out` → FAIL).
 - EV-T6.2 §9.2–9.5 integration/E2E/stress/chaos suites + a challenges Challenge asserting the event→delta < 1 s SLO with captured evidence (§11.4.27/§11.4.69/§11.4.107).
- **EV-T7 — Phase-2 seam proof.**
 - EV-T7.1 Bus interface conformance test-suite re-runnable against any impl (the seam proof for the NATS swap, §10.2).

12. Edge cases (explicit, no-guessing — §11.4.6)

#	Edge case	Defined behaviour
E1	XADD succeeds but the relay crashes before <code>MarkOutboxRelayed</code>	Row stays pending → re-relayed → duplicate XADD → idempotent reaction absorbs it (§6.2). At-least-once, never lost.
E2	Reaction partially completes then errors (e.g. pushed delta to 3 of 5 streams)	Not acked → reclaimed → re-reacted; recompute-from-state re-pushes all 5 (the 3 already-correct are no-op diffs). Safe.
E3	Duplicate <code>policy.compiled</code> for the same version	<code>shouldApply</code> drops the second (version ≤ applied marker). No double recompute.
E4	Out-of-order delivery: v=6 reclaimed <i>after</i> v=7 applied	Monotonic guard drops v=6 (no regression). The taxonomy uses <code>version/gen</code> , not wall-clock, exactly for this.
E5	Redis fully flushed while <code>helixd</code> up	Coordinator graph already current from PG (C2); next mutation re-publishes via outbox; PEL reset is harmless.
E6	Poison event that NO consumer can ever process	Redelivered <code>ceiling=5</code> times → DLQ + alert; operator inspects/fixes/replay-dlq. Never spins forever (§6.5).
E7	Undecodable message (corrupt fields)	Straight to DLQ on first delivery (no retry loop on a structurally-broken message), <code>dlq_reason="bad ..."</code> .
E8	Producer clock skew	Ordering keys are monotonic counters, not ts; ts is informational only — skew cannot misorder topology changes (§9.5 chaos).

#	Edge case	Defined behaviour
E9	Slow audit/ws-fanout consumer	Distinct group/PEL — cannot stall coordinator; isolated back-pressure (§6.6).
E10	Event for a tenant whose RLS context is unset	guardPublish rejects a non-system event with <code>tenant_id == Nil</code> ; consumer fan-out is tenant-scoped so a malformed-tenant event cannot leak across tenants (C4).
E11	DLQ itself grows unbounded	DLQ MAXLEN $\sim 1_000_000$ + nightly audit export; alert fires long before the cap.

13. Sources

[02_CP] final/02-control-plane.md (events backbone §5, coordinator §6, SLO §10, invariants §0.1, RLS §2.3, proto `helix.coordinator.v1` §4) · [04_P1] HelixVPN-Phase1-MVP.md (Redis Streams backbone, at-least-once + idempotent reactions, DLQ, $p_{99} < 1$ s convergence, Phase-2 NATS seam §12) · [04_ARCH §4] HelixVPN-Architecture-Refined.md (schema-generated clients, control/data separation, no-logging, default-deny) · [research-go_cp] cited research (accessed 2026-06-25): Redis Streams consumer-groups + XAUTOCLAIM dead-letter + delivery-count ceiling + Redis 8.4 CLAIM; `redis/go-redis/v9`; transactional-outbox async path; goose RLS migrations — full source list with access dates in that file's "Sources verified" section · [SYNTHESIS] §2 (stack floor), §3 (D3 event-bus decision Redis→NATS), §7 (no-logging invariant), §8/§9 (ecosystem + constitution bindings).